CS2240 Final Project Proposal

The Turtle Club

Gavin Dhanda, Ryan Aubrey, and Briana Fedkiw

Paper:

- Going with the Flow — Yousuf Soliman et al., 2024
  - PDF: (https://www.yousufsoliman.com/projects/download/GoingWithTheFlow.pdf)
  - Website: (https://www.yousufsoliman.com/projects/going-with-the-flow.html)
  - Demo: (https://www.youtube.com/watch?v=iJPIT_BsSVA)
- We plan to implement the entirety of this paper scoped to a single demo of the turtle. This includes Precise pose sequence for demo, a time-varying inertia setup, generalized added mass of background flow, localized lift and drag forces, and more.

Extensions:

- We want to address a limitation mentioned in Section 7.5 of the paper. Using one global scaling factor for added mass. In the original method, the paper computes added mass locally, then applies a body-level geometric normalization based on average mean curvature. This works well for simple shapes, but it can be very coarse for more complex bodies. Our extension is to split the mesh into several parts, compute a separate geometric scale for each part, and combine those into a piecewise added-mass approximation as recommended in the section. We will compare this against the original global version and test whether it improves accuracy. We can compare the two trajectories specifically.

Resources:

- SideFX's Houdini (to run the simulation) - There is a noncommercial version that is free and will not require anything; it will simply be limited to 1920x1080 resolution and be watermarked.
- Maya - Modeling tool for our single demo. This is not free software but Briana already owns a subscription.
  - Possibly buy a model instead
  - https://www.turbosquid.com/3d-models/turtle-679632

Rough Timeline:

Note: Below timeline is subject to change based on our understanding of the paper
Days until presentation: 38
Weeks to work: 5 weeks

..	**Date Done By**
Turtle model + rigging rough draft (via Maya) OR buy model and save animation into separate poses	4/8
Integrating the model into code (mesh loader for vertices, faces, etc)	4/10

Implement time-varying inertia and added mass	4/10
Calculate lift & drag to get total force	4/12
Implement the other motion equations (as functions that can be called during discretization?)	4/16
Calculating the kinetic energy of the body & fluid	4/16
Discretization of motion equations (see algo in paper)	4/18
Validate/Debug with simple model (take simple open source object or turtle)	4/20
Extension implementation	4/26
Compare global added mass with piecewise added mass	4/26
Final rendering (Houdini)	4/30
Presentation slides & prep (answering the questions in the doc)	5/4
Presentation Day	5/8
Finalize code & GitHub (README)	5/10
Materials Due	5/11

Division of Work:

Gavin	**Ryan**	**Briana**
• Calculate lift and drag to get total force • Implement the motion equations • Extension implementation/ testing • Presentation Slides/Github	• Implement time-varying inertia and added mass • Validate/Debug with simple model • Extension implementation/ testing • Final Rendering • Presentation Slides/Github	• Turtle model • Integrating the model into code • Kinetic Energy calculation • Discretizing the algo (combining the equations) • Extension implementation/ testing • Presentation Slides/Github

Notes:

- add a dependency chart, may lead to changes in division of work/timeline
- allow time to work through math and understanding it